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Reconstructing Aura in immersive heritage media

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1. Introduction & Research aim

Museums are gradually shifting away from their traditional role of merely gathering and disseminating knowledge and becoming places where peop le can experience and interact with cultural heritage (Recent discussions at the International Council of Museums, 2023). This paradigm shift redefines museums as creators of interpretative narratives rather than merely keepers of artefacts. By converting historical materials into emotionally engaging, spatial experiences, immersive media technologies, such as augmented reality (AR), projection mapping and interactive installations, reaffirm institutional authority in this context. In particular, immersion media enhance experiential elements and introduce 'infotainment' innovations into heritage representation, creatively depicting cultural heritage that is difficult to reconstruct or only partially documented (Tschumi, 2021). These new mediation techniques are highly regarded for their capacity to vividly reconstruct lost architectures, rituals and traumatic events and to create deep emotional connections to the past (Hermund et al., 2018). Nevertheless, despite continuous technological and paradigmatic changes, museums' institutional duty to preserve and disseminate the inherent value of cultural heritage endures (Bevan, 2020). The use of immersive media to represent heritage therefore raises the following research question: Does immersive media simplify and reduce these meanings to consumable images, or can it fully convey the original's historical depth, cultural context and artistic value?

A key conceptual framework for understanding the significance of today's digitally mediated artistic works is Walter Benjamin's idea of the 'aura' of artworks (Benjamin and Arendt, 1935; Sweeney, 2019). Benjamin specifically describes aura as an object's distinct presence firmly anchored in its historical setting. However, the reconstruction of cultural artefacts using immersive media can result in what Jean Baudrillard refers to as 'hyperreality' (Benjamin, 1977), a simulation that is more captivating than reality. This can provide an even more vivid sensory experience than the original. In this study, we situate immersive media within a long tradition of spectacular displays, from 19th-century battle panoramas to modern virtual reality (VR) installations. This approach emphasises the critical roles of mediation, audience subjectivity, and ideological structures in understanding digital reconstructions of historical representations. Finally, we demonstrate the effectiveness of our proposed critical methodology and explore practical alternatives by examining several case studies of immersive cultural heritage. Therefore, the aim of this research is to critically investigate how immersive media technologies reconstruct aura, authenticity, and cultural memory within cultural heritage representation.

2. Theoretical framework and literature review

Before we can achieve the objectives of this research, it is necessary to understand the critical foundation upon which this paper is based. To deeply understand the artistic, philosophical and ideological implications of immersive media, we will examine key concepts presented by leading theorists such as Benjamin, Baudrillard, Hall, and Asman.

2.1. Literature review

Walter Benjamin argues that historical objects or locations have an aura as originals (Grau, 2003a). According to Benjamin, cultural artefacts — even ruins with little remaining evidence of their age — exist as allegories of the originals, which can be recreated using countless texts. Allegorical interpretation gathers pieces of the past to create new meanings. This study addresses critical questions of whether heritage remediated through immersive media can allegorically reconnect with the original.

Jean Baudrillard argues that hyperreality results from signs representing other signs rather than reality in the era of simulation. From this perspective, it is unclear whether immersive media's hyperreal heritage representations allow viewers to access historical facts or merely provide improved simulations that distort perceptions. Immersive media simulations of heritage can turn into hyperreal spectacles. Who is in charge of this delusion? These enquiries have a direct bearing on the

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encoding and ideological concerns the following literature covers.

Stuart Hall contends that the interactions among the encoder, the text and the decoder — which are heavily impacted by social and cultural contexts — create the media's meaning. When applied to the representation of cultural heritage, it becomes evident that those designing immersive experiences invariably incorporate specific historical narratives and presumptions. Therefore, rather than viewing immersive media as neutral windows, Hall's framework encourages us to view them as 'representations' susceptible to biases and interpretations. With the ability to emphasise certain interpretations — such as visualising particular heritage theories or highlighting specific historical figures — and suppress alternative possibilities, immersive media emerges as a new medium for cultural construction. This realisation is essential when debating whether immersive representation promotes pluralistic storytelling or serves to reinforce dominant narratives.

Jan Assmann highlights that 'institutionalised memory techniques', developed top-down by authorities, are used to preserve cultural memory. Museums, archives, memorial services and educational initiatives purposefully shape the preservation of memories. Such practices give members of society continuity and a common identity based on historical narratives. Just as photography and film forced museums to re-evaluate authenticity, the current paradigm shift towards immersive media compels organisations to reconsider their significant 'connections' with the past.

The four theoretical perspectives outlined above enable immersive digital heritage to be understood not as a simple binary of "real/false," but as a field in which experience, memory, and power intersect. Benjamin's concept of aura first asks whether immersive media can reconfigure, in the absence of a physical original, a form of "experienced presence at a distance," or whether it instead effaces precisely that distance. In response, Baudrillard's discussion of simulacra and hyperreality highlights the danger that immersive heritage may construct an image-world whose sensory appeal and persuasive force exceed those of actual history, making it difficult for viewers to distinguish the vividness of their experience from historical factuality. Taken together, these two positions mark the opposite poles of, on the one hand, the possibility of aura and, on the other, the risk of hyperreal distortion, and thus constitute a tension that must necessarily be held in view when analyzing immersive heritage. Accordingly, in this article the four theorists are not treated as separate explanatory frameworks, but as complementary axes for the analysis of immersive cultural heritage. Benjamin–Baudrillard provide an aesthetic and sensory dimension in terms of aura/hyperreality, while Hall–Assmann provide a social and institutional dimension in terms of encoding and practices of cultural memory. In the chapters that follow, these four axes are used to compare and analyze the extent to which nineteenth-century panoramas and contemporary VR-based heritage projects simultaneously invoke aura and produce hyperreal spectacle, and how, in the process, they encode and reconfigure historical narratives and cultural memory.

3. Materials and methods

3.1. Case study design and selection rationale

This study uses a qualitative case study approach in order to examine how the theoretical issues presented earlier operate in practice. Based on academic relevance, historical significance and technological innovation, two immersive heritage projects were selected as case studies.

3.1.1. Suriname flag VR exhibition (2024)

This experimental VR installation explicitly presents multiple perspectives on the flag—an emblematic discourse of national ideology—by focusing on polyvocality as an alternative to the media-specific problems of immersive cultural heritage, namely the imposition of a single authoritative narrative, artificial authenticity and the passive role of the user (Franke et al., 2024a). Developed by researchers at Vrije Universiteit Amsterdam, the work curates the history and meanings of the Surinamese flag since 1975 through the perspectives of various communities and explicitly investigates how conflicting narratives (Indigenous, immigrant, colonial, etc.) can be incorporated into design instead of a single authoritative storyline (Franke et al., 2024a).

3.1.2. The Lost Palace (2016)

In Benjamin's well-known formulation, aura is bound up with the "appearance of distance" that remains even when an object is physically close. Here, "distance" refers not only to a physical gap but also to the temporal, spatial and ritual interval that the viewer can never fully possess. The second work selected, The Lost Palace, connects with 'traces of the past' solely through auditory and bodily perception using binaural sound and haptic feedback, maintaining non-visibility instead of filling the palace's absence with compensatory visual reconstruction (Allen et al., 2018; Calvium, 2018). This strategy intentionally leaves a gap between the present (the contemporary Whitehall streetscape) and the non-present (the vanished palace), allowing the viewer to imaginatively fill that interval (Allen et al., 2018; Calvium, 2018). This nearness-within-distance corresponds to the experiential condition of aura described by Benjamin and, for this reason, the work was selected as a counterpoint to the visually driven VR case study (Allen et al., 2018).

3.2. Immersive media development process

The two immersive cultural heritage cases can be summarised in four stages: 3D data collection and modelling, production platform and software stack, metadata schema and content documentation and interaction design and narrative scripting.

3.2.1. 3D data acquisition and modelling

The digital assets of the Suriname Flag VR are classified into (a) symbolic narrative sources belonging to the flag (colours, iconography and historical events), (b) 3D resources for constructing the historical

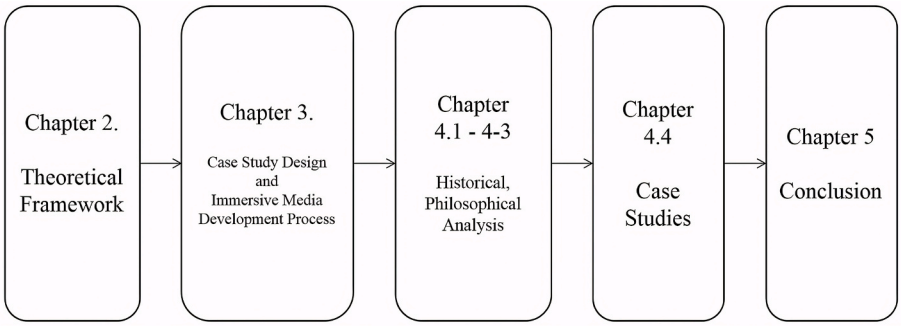

Fig. 1. Overview of the research design and analysis pipeline.

scenes and (c) voice and sound resources, which were collected and produced accordingly (Franke et al., 2024a). Virtual environments and objects were modelled using Blender and Rhino 3D, and the natural terrain for the Green Sphere scene was created within Unity using a terrain-generation workflow documented by the project team (Franke et al., 2024a). Character representation was implemented using avatar animations created with the iClone/Reallusion pipeline and imported into Unity as animated FBX assets (Franke et al., 2024a). Voices were synthesised through Replica Studio to design scene-specific narration tones (neutral/contrast/testimony), and a scanned 3D object of the flag, provided by a museum partner, was used to match the form and texture of the symbol (Franke et al., 2023a, 2024a).

For The Lost Palace, the primary assets consist of audio and haptic materials rather than 3D models. The sound assets were organised into three categories: (a) narrative and character voices that drive the story; (b) environmental and ambient sounds that form the spatiotemporal atmosphere; and (c) interaction-linked sound effects and haptic-synchronised audio (Allen et al., 2018; Calvium, 2018). Episode-specific dialogue scripts were written on the basis of historical narratives related to Whitehall Palace and recorded with voice actors in a studio, then mixed into binaural (3D) audio for headphone listening so that the position, distance and direction of characters could be perceived auditorily (Allen et al., 2018; Calvium, 2018). Contemporary urban noises of Whitehall and recreated historical environmental sounds (court banquets, crowds, fire, bells, etc.) were produced as separate ambient tracks and layered so that the acoustic strata of the past overlap with the real urban space through which visitors walk (Allen et al., 2018; Calvium, 2018).

3.2.2. *Production platform and software stack*

The Suriname Flag VR was developed in Unity (2021).3.17 LTS, using the XR Plug-in and SteamVR stack to integrate the HTC Vive Pro HMD and controllers (Franke et al., 2024a). Scene-specific LOD and occlusion culling were applied for real-time rendering optimisation, and the overall system was tuned for stable framerates on a lab-based VR setup (Franke et al., 2024a). The soundscape for each scene was constructed by mixing stereo and binaural layers, whereas timing and interaction events were managed through C# script-based state machines (Franke et al., 2024a). Development and testing were conducted in a laboratory environment, and in consideration of the wired constraints of the HMD, teleportation and room-scale movement were combined and interaction ranges were reduced (Franke et al., 2024a).

The Lost Palace runs on a custom handheld device embedding an Android smartphone (developed by Calvium), and the app integrates sensor triggers, timed triggers and NFC reading triggers to control multiple audio tracks simultaneously (Allen et al., 2018; Calvium, 2018). The smartphone connects with an external circuit board that drives haptics using an audio transducer and provides auxiliary power (Allen et al., 2018; Calvium, 2018). GPS is used to trigger situation-based background tracks and gyroscope/accelerometer sensors are used to trigger gesture sounds. The NFC reader is repositioned to the top of the device so that visitors initiate scene entry by 'placing' the top of the device onto blackened wooden markers along the route (Allen et al., 2018; Calvium, 2018).

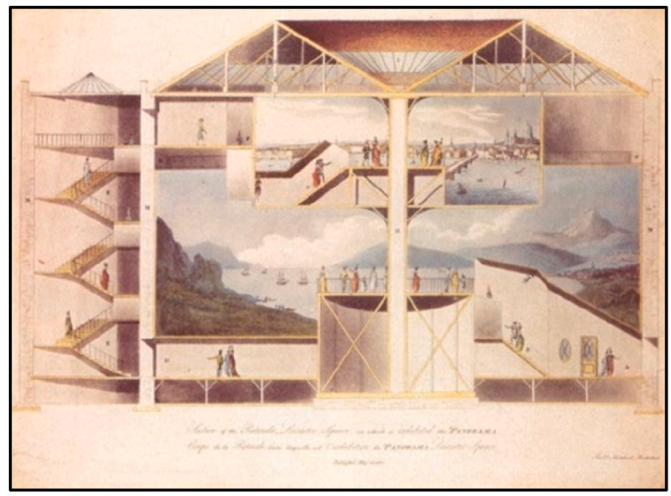

Fig. 3. Panoramic view of London, sectional view, aquatint, 28.5 x 44.5 cm. Gebr (Barker, 1792).

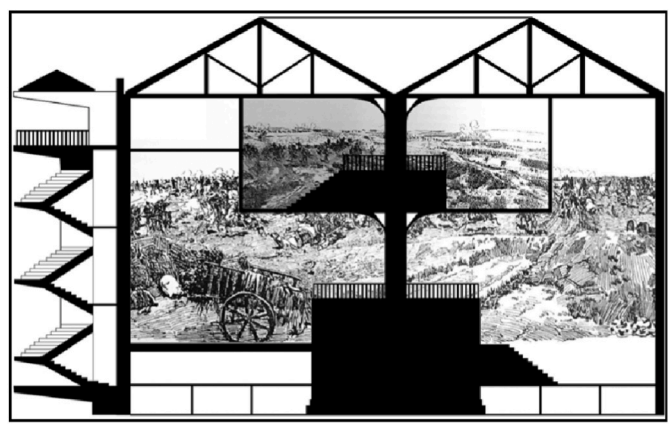

Fig. 4. Collage drawing of the circular panoramic hall with images from the Battle of Sedan Panorama.

3.2.3. *Metadata schema and content documentation*

Suriname Flag VR. All design, content and evaluation materials for this project are publicly archived on the Open Science Framework (OSF), including the exhibition walkthrough video, survey instruments, interview transcripts and procedural documentation (Franke et al., 2023a). Based on the organisational structure of the OSF repository, this study generated a scene/object-level content registry detailing: provenance sources (archival records, community interviews and the scanned flag object); descriptive intent and symbolic interpretation; production method (3D modelling, photogrammetry and synthetic voice generation); and interaction events (trigger conditions and feedback) (Franke et al., 2023a, 2024a). This metadata field structure provides the minimal technical unit necessary for subsequent analyses of aura, polyvocality and hyperreality at the scene level and supports data reuse and reproducibility for future research (Franke et al., 2023a, 2024a).

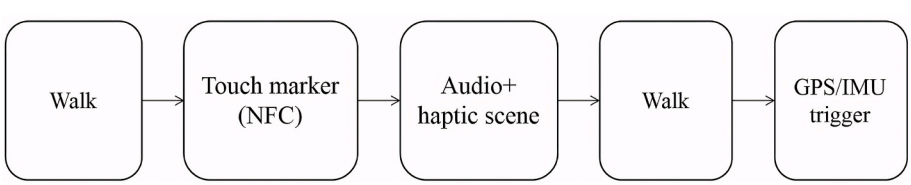

Fig. 2. User journey and interaction loop in The Lost Palace.

The Lost Palace. The original audio files, dialogue scripts, sound-design session materials and other core production assets are subject to multiple copyright holders and contractual restrictions, and therefore cannot be deposited in an open repository such as the OSF (Allen et al., 2018; Calvium, 2018). Accordingly, this study adopts an approach that reconstructs only the 'minimum reproducible layer' of metadata ex post facto. Based on information verifiable through developer correspondence and publicly available project descriptions, we compiled a concise technical summary table (Table 1) that clarifies the structural relationships among the platform (hardware–software coupling), route–marker mappings and user experience log (Allen et al., 2018; Calvium, 2018; Historic Royal Palaces, 2016).

3.2.4. *Interaction design and narrative scripting*

Suriname Flag VR. The central goal of the Suriname Flag VR is the experiential learning of polyvocality (Franke et al., 2024a). In the central pavilion, users freely select the red, white and green spheres, transitioning into sub-environments connected to their symbolic meanings (Franke et al., 2024a). In the Red scene, the meanings of oppression/-struggle and love/progress are juxtaposed through book objects and heartbeat rhythm sounds; in the White scene, independence and pre-independence flags are presented through a two-avatar dialogue alongside parallel visual materials, rendering coexisting conflicting perspectives; and in the Green scene, an immersive natural soundscape and realistic visuals are followed by falling sacks representing agricultural products, evoking economic–political layers. After completing the three spheres, the Star scene removes visual stimuli (blackout) and has users listen to community testimony to induce attentional focus and self-reflection. The overall sequence returns to Room 1 through Ring-komposition, and interactions are managed through a state machine (Franke et al., 2024a).

The Lost Palace. The experience is structured around on-site movement and the ritual of contacting markers. According to the headphone guide's prompts, the visitor touches the top of the device to a black wooden marker (with embedded NFC tags) in order to enter a historical audio scene (Allen et al., 2018; Calvium, 2018). GPS triggers location-based background layers and the gyroscope/accelerometer triggers gesture sounds (Allen et al., 2018; Calvium, 2018). The app synchronises multiple audio tracks with haptic (audio-transducer) feedback to provide combined auditory–tactile sensations. The narrative is constructed by scattering fragmentary episodes (snippets of times, people and events) throughout the site, designed so that users co-construct meaning through repeated cycles of walking, pausing and contact. By excluding full visual reconstruction, the experience avoids hyperreal saturation and preserves room for distance, imagination and interpretation (Allen et al., 2018; Calvium, 2018). In the following Chapter 4, building on these implementation details as a point of departure, the analysis will broaden its focus to a historical and theoretical scale in order to examine the continuities and discontinuities between nineteenth-century battle panoramas and contemporary VR-based heritage spectacles, and—by applying the conceptual frameworks of Benjamin, Baudrillard, Hall, and Assmann outlined in Section 2—critically investigate how immersive media invoke or efface aura, and how, in the process, they encode and reconfigure historical narratives and cultural memory.

4. Results and discussion

4.1. *Historical analysis: panoramas and aura*

To effectively understand the implications of contemporary immersive media such as VR and AR in reconstructing historical experiences, it is crucial to first examine earlier historical forms of immersive media and their strategies of artificial historical reconstruction. In this chapter, we will examine how panoramas, the result of immersive image strategies developed in the 19th century, artificially process the aura of historical time.

Oliver Grau considers enclosed image spaces important sources in media studies when defining the aesthetic qualities of technology-based media, such as VR, AR and transcendental media (Oettermann, 1997a). These are somewhat ambiguously conceptualised as 'spaces created by completely hermetically enclosed images, fully separated from the external world'. Mostly created in 19th-century Europe, panoramas are examples of image space that incorporate the most advanced optical technologies available at the time. According to Grau, they are among the earliest examples in the history of media that use techniques to completely engross viewers in images to convey messages. Today's VR recreates historical auras in contexts formed by critically examining how the visually represented time of panoramas allegorically relates to the original aura of historical events.

To observe the large cylindrical space, which was designed to maintain a fixed distance between viewers and the immersive imagery, viewers had to climb stairs to reach the central viewing platform (Werner, 1883). Sunlight was able to continuously illuminate the paintings on the walls through side and ceiling openings, creating amazing effects that enhanced the interior. A well-known example is the Battle of Sedan Panorama(1883) in Berlin, which depicts Prussia's victory over France in 1870 (Mitchell, 1801). Having just been united by war, Germany used the triumphant portrayal of this panorama in the 1880s to reinforce its national identity through heroic memory.

Historical documents from Prussian military archives were recreated meticulously under Werner's supervision to capture the tragic scene of the battlefield on 1 September 1870, close to Alexander Hill, where Prussian and French troops were engaged in combat (Oettermann, 1997b). As if surveying the battlefield like commanders, viewers standing on the elevated platform have a 360-degree view of the terrain. The artwork depicts French soldiers retreating and attempting to flee as Prussian troops encircle them. Amidst the chaos, the French cavalry is scattered as the well-organised and determined Prussian forces pursue the retreating French soldiers. Oliver Grau points out that the panorama only showed the victors' perspective although the Battle of Sedan featured Prussian massacres of French civilians (Grau, 1997). The aura the Sedan panorama creates shifts and changes the aura of the actual historical event of the 1870 battle, serving as an example of a new aura infused with the idea of the modern 'nation', which was essential to the developing German national identity. By evoking strong emotional investment in national history, producing the illusion of firsthand experience and reducing the gap between viewers and events through visual

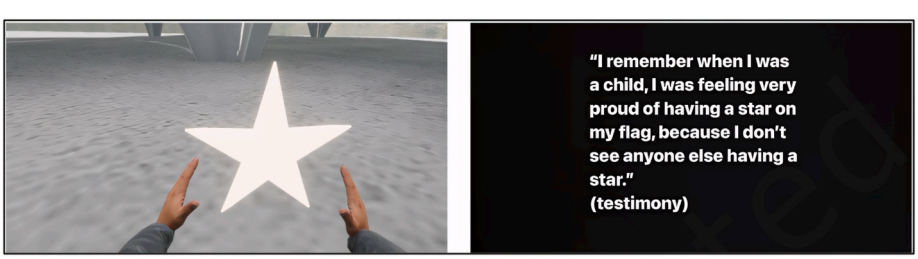

Fig. 5. Installation view from *Flag of Suriname VR*, a polyvocal virtual heritage exhibition developed by Franke et al. (2024) (Franke et al., 2024b).

immersion, panoramas are an early example of immersive media used for nation-building.

Panoramas' intense sensory realism and meticulous visual consistency lead audiences towards prevailing interpretations, severely limiting interpretive flexibility. The Sedan panorama employed realistic details and visual immersion to reinforce a singular narrative of courage and national solidarity. As the 360-degree illusion saturated viewers' perceptions with simulated reality, opportunities for critical evaluation were diminished. Hall's encoding concept is therefore essential in this immersive setting. Panoramas effectively limited interpretive autonomy, forcing audiences to accept encoded nationalist messages as unquestionable truths and defining particular ideological interpretations as objective historical realities. Thus, panorama media exhibit an elevated form of ideological encoding, whereby sensory immersion limits critical or opposing readings while reinforcing dominant interpretations.

4.2. Virtual reality and the spectacle of heritage

Through visual immersion, 19th-century panoramas successfully conveyed particular historical narratives and ideological messages to viewers, as the Battle of Sedan demonstrates. By offering complete experiences that incorporate visual representation, sound, touch and interaction, modern VR builds on these strategies to create an even more immersive experience. However, this enhanced realism and sensory immersion may not always fully capture the original's historical accuracy and artistic depth. In fact, it can mislead viewers into believing that mediated experiences are more realistic and reliable than reality. possibly causing them to unquestioningly accept certain ideological interpretations supported by governments or institutional authorities, just as panoramas did in the past.

To increase credibility and the sense of experiencing something ‘as it really was’, VR aims for what Bolter and Grusin refer to as ‘immediacy’ (Bolter and Grusin, 1999), making users feel as though they are ‘actually there’. Ironically, this hypermediality — enhancement of media elements — creates a sense of immediacy or transparency for the user. Users can experience direct presence in historical locations or events through the skilful integration of VR's multiple mediation layers, including digital models, game engines, headsets and sensors. The paradox of hypermediality is that the experience seems more ‘transparent’ the more thorough and seamless the mediation. By portraying the created reality as firsthand observation, this effect can subtly entrench preferred interpretations and greatly reinforce mainstream narratives in heritage simulations. Media features that fix viewers' gaze within image spaces completely enclose them, blurring the distinction between the external and internal worlds, as Oliver Grau points out. This reduces viewers' awareness of artificiality, making them believe that the VR experience is real (Grau, 2003b).

Hypermediality easily leads viewers towards prevailing interpretations by creating the illusion of transparency and immediacy, turning historical reality into consumable spectacles. Guy Debord's concept of the ‘spectacle’ (Debord, 1995a) is closely linked to the immersive immediacy of VR in this context, for it dissolves historical facts and social relationships into illusory images. Consequently, historical representations become spectacles due to the ideological impact of the media's quest for immediacy. He argues that these spectacles mediate social relationships because individuals consume identical packaged historical images instead of engaging in critical dialogue as a group. Because it is impossible to interact with historical processes,

Fig. 7. Still from *The Lost Palace: Bringing History to Life Through Augmented Reality* (Calvium, 2016) (Calvium, 2016).

Table 1
System architecture of the Suriname Flag VR application.

Input	Engine	Output
OSF-based 3D models, sound assets, avatar animations	Unity 2021.3.17 LTS + XR Plug-in/SteamVR + state-machine interaction	HTC Vive Pro HMD + binaural audio

Table 2
Summary of reconstructed metadata elements for the lost palace (Historic Royal Palaces, 2016).

Category	Elements (Reconstructed)	Source/Certainty
Hardware–Software Coupling	Android app; NFC reader (top-mounted); GPS; gyroscope/accelerometer; external PCB controlling audio–haptics; bone-conduction headphones	Developer correspondence (high) (Allen et al., 2018; Calvium, 2018)
Route–Marker Mapping	Blackened wooden markers (archway, post) embedded with NFC tags; scene entry triggered by touching marker; GPS zones triggering background layers	Developer correspondence + field reconstruction (medium) (Allen et al., 2018; Calvium, 2018)
User Experience Log	Visitor comments coded for presence, emotional response, confusion, historical reflection; spreadsheet transformed into structured log	Guestbook dataset (high) (Historic Royal Palaces, 2016)

Fig. 6. Installation view from *Flag of Suriname VR*, a polyvocal virtual heritage exhibition developed by Franke et al. (2024).

nations and museums that produce spectacles use them as a tool for unification and separation. Furthermore, the objective of immersive spectacles in heritage contexts is often to 'auratise' history and inspire reverence for the nation's past. In this case, Debord's observation that 'only illusion is sacred and truth profane' applies because painstakingly altered and romanticised historical simulations have the power to inspire sacred sentiments regarding heritage while obscuring the more nuanced and less glorious realities of history (Debord, 1995b). Consequently, spectacles supported by the state serve to legitimise current social structures.

Raymond Williams's concepts of 'dominant', 'residual' and 'emergent' provide a helpful theoretical framework for discussing how immersive media support historical narratives and investigate alternative, critical cultural narratives that might emerge through new media forms, such as VR (Williams, 1977a). Williams uses the term 'dominant' to describe how dominant cultural norms and ideologies influence our perception of the world and our knowledge (Williams, 1977b). This is evident in the preservation of popular historical and cultural narratives in immersive VR experiences. As discussed in the previous chapter, the panoramas of the 19th century can be regarded as a dominant medium that conveys '*the implicit power of sovereign states* (Grau, 2003c),' as Grau stated. VR experiences at museums or historic sites often reinforce existing historical interpretations that align with prevailing ideologies, thereby maintaining established power structures and marginalising alternative perspectives. Williams's 'residual' elements, rooted in historical customs or cultural forms inconsistent with current ideologies, exemplify past influences still impacting the present (Williams, 1977c). These residual elements may manifest in immersive VR through conventional storytelling or the incorporation of historical artefacts, thereby conserving heritage components and promoting historical narrative negotiation and contemplation. New cultural forms and ideologies resulting from shifting social conditions are referred to as 'emerging' elements. Immersive VR recreates historical events in a creative way while also questioning conventional expression techniques. This emergent quality has the power to elevate under-represented voices and histories, thereby transforming audiences' perceptions of and interactions with the past. If so, what practical strategies should be adopted to enable immersive media to fulfil its role as an emerging medium and transcend its current marginal position? This question will be explored in depth in the next section.

4.3. Reconstructing aura through digital interaction

According to constructivist approaches in media theory and museology, aura is socially negotiated among contexts, objects and observers (Sitzia, 2017). According to Bruno Latour and Adam Lowe's famous 'migration of aura' theory, aura can move from originals to high-fidelity reproductions (Latour and Lowe, 2010; Hein, 1998). They argue that expertly crafted replicas can augment the original's aura, enabling its significance to be appreciated in novel contexts and by diverse audiences. The preservation of significant components — forms, details and contexts — in new media is essential to this migration. According to Notarangelo, cutting-edge methods such as photogrammetry and 3D reconstruction can enhance conceptual aura by strengthening ties to cultural heritage, thereby demonstrating the capacity of technology to connect with audiences in a genuine way (Notarangelo, 2019). Stuart Jeffrey builds on this idea and suggests that under certain conditions, digital replicas can possess an aura (Jeffrey, 2015a). To enable aura in 3D models and virtual reconstructions, he emphasises the importance of coproduction, involving multiple stakeholders in the creation process, content democratisation and aesthetic considerations. When carefully crafted and presented, virtual heritage experiences can evoke feelings akin to those experienced with original artefacts. With this redefinition, aura is no longer considered unique to originals but rather an emergent property resulting from interactions between viewers and digital objects. This concept is supported by Alison Landsberg's idea of 'prosthetic memory', which suggests that media and technology can internalise historical experiences and promote moral responsibility for the histories of others (Landsberg, 2004). Through sensory representation, immersive VR has the potential to embody past traumatic or suppressed experiences, urging users to internalise them as personal memories and encouraging critical reflection on prevailing power structures.

Jay David Bolter's viewpoint on digital aura further supports this, contending that rather than eliminating aura, new media technologies have placed it in a state of flux, making it possible at all times (Bolter, 2019). He argues that by focusing on performativity to achieve authenticity at experience moments and by relying more on viewer participation and context than on material origins, designers can evoke aura in novel ways through the arrangement and presentation of digital content. As Poster and Jones suggest, digital objects gain aura and authenticity over time through context, use and interpretation (Poster and Jones, 2009). By arguing that authenticity is experiential and co-created, this viewpoint directly challenges the idea that VR is fundamentally misleading. However, there are some potential drawbacks to the migration or replication of aura through immersive media. According to Stuart Jeffrey, the unique qualities of digital objects — such as immateriality, freedom from deterioration, infinite reproducibility and ambiguous provenance — may separate virtual representations from historical and sociocultural contexts, reducing their aura. Without context, virtual models could appear weightless or out of place. Recognising this hazard highlights the importance of making up for physical absence with carefully chosen methods and designs that connect the digital and physical worlds. Consequently, aura does not simply disappear in virtual environments; rather, it manifests when digital experiences are created in a way that promotes interpersonal relationships. According to Rokeby, interactive systems are 'transformative mirrors' that enable viewers to examine and consider cultural heritage creatively (Rokeby and Dodsworth, 1998a). Through gesture-based interactions, digital interactivity enables users to engage directly with cultural narratives, thereby fostering empathy and introspection. Stuart Jeffrey argues that providing users with interactive control restores the critical distance that pure immersion eliminates while fostering connectivity and authenticity by enabling users to explore their curiosity and interpretations.

By incorporating viewpoints that would otherwise be excluded, immersive media serve as platforms for convergent voices, thereby reducing ideological biases. Sarah Kenderdine highlights that interaction is essential to creating meaningful connections in digital heritage environments (Kenderdine, 2016). Viewers participate in creating content by sharing their memories, questions and interpretations. By providing spaces for active remembrance, reinterpretation and historical engagement, VR installations act as new symbolic vessels for cultural memory, augmenting traditional monuments and texts. Therefore, the concept of 'aura' is not confined to original artefacts but rather spreads among participants in digital environments. This concept resonates with Latour and Lowe's idea of a migratory aura because it emerges from collective memory processes and can move across communities, platforms and locations. Unlike closed ideological exhibits, immersion media function as an open forum by encouraging interaction.

4.4. Case studies in multi-voiced heritage practice

In the previous sections, we examined historical perspectives and critical frameworks of immersive heritage, focusing on three main issues:

(1) Immersive VR imposes singular, ideologically closed narratives, also known as 'ideologically closed image spaces'.
(2) The authenticity aura it creates is artificial or potentially deceptive.
(3) Users' roles are reduced to passively accepting curatorial viewpoints.

The following methods are proposed as ways to mitigate or overcome the conceptual and experiential issues discussed above when using immersive cultural heritage media.

1) Multiple narratives: Immersive experiences can incorporate multiple narratives and perspectives rather than presenting a single, officially approved history. When conflicting or varied perspectives are included in the same virtual space, users must interact with and reconcile divergent points of view, which prevents design closure. This multi-voiced approach keeps image spaces open for discussion and debate by ensuring that no one ideology fully controls interpretation.
2) Interactive agency: This prevents ideological imprinting by giving users the freedom to explore, choose or change the experience. Users participate in forming the narrative when they select stories or explore their own interests. As users seek out content rather than being told what to think or feel, interaction fosters critical engagement. According to research, well-designed interactivity creates a personalised experience by adjusting to user input (Blasco-Arcas et al., 2014). Consequently, interactive design transforms immersive media into spaces for two-way communication rather than passive consumption.
(3) Contextualisation and transparency: Rich contextualisation of immersive content is one of the most effective ways to address the 'closed image' issue. Users can understand what they see and why it appears as it does by placing VR within a framework of sources, explanations and acknowledgements of uncertainty. Heritage theorists emphasise that context is the primary source of authenticity and meaning for models and replicas (Chapagain, 2017). When connected to familiar stories or locations, digitally recreated objects take on a similar sense of meaning to real exhibits when placed in context through interactive guides, virtual signs, audio commentary, etc. It is equally important to be transparent about the reconstruction process, for example by clearly identifying interpretative variability points and hypothetical elements. Curators encourage users to maintain critical awareness while immersed by openly acknowledging artificial elements. As a result of this transparency, closed illusions become self-aware and instructive investigations.

VR experiences that are multi-voiced, interactive, co-created and transparently contextualised oppose spaces that are ideologically closed. These tactics generate new areas of meaning rather than obstructing interpretation. Because users perceive immersive experiences as artificial representations that encourage inquiry and investigation rather than offering unbiased windows into reality, these experiences can become part of the narrative.

4.4.1. *The Surinamese Flag VR project (2024)*

We examine a recent VR project that curates multiple narratives around a single heritage object, demonstrating how immersive design incorporates diverse voices and promotes thoughtful interpretation. The experimental VR exhibition featuring the national flag of the former Dutch colony of Suriname is a noteworthy example of this. The Surinamese Flag Project was created by researchers at Vrije Universiteit Amsterdam to examine the Surinamese flag's cultural and historical significance from a variety of angles (Franke et al., 2023b). In this VR experience, users are given access to multiple virtual spaces, which illustrate how various communities, such as immigrants, indigenous peoples and independence activists, interpret the flag's colours and symbols differently. Through their actions and decisions, users activate different scenes, creating unique experiential pathways.

To combat colonial biases in museum collections, this project presents the history of the Surinamese flag from various perspectives. To understand the flag's colours and symbols from various historical and cultural viewpoints, participants investigated three virtual scenes. For example, one viewpoint linked the red and green of the flag to Surinamese independence leaders, and another explained the significance of these colours in indigenous and immigrant communities. Users essentially had to switch reference frames while navigating these scenes, comparing and balancing various interpretations of the flag's meaning.

As participants reported, the flag, and by extension any heritage object, has multiple meanings rather than a single authoritative narrative. According to empirical research, the VR exhibition encouraged viewers to adopt a 'multi-perspective mindset' (Franke et al., 2024c). Users became more open to different interpretations and more aware that the meanings of artefacts and symbols can change depending on the context and perspective. Due to the VR system's adaptive design, users could select from various content sequences, providing them with some degree of control over their experience. One of the main elements of the multi-perspective approach was this shared authorship. By the end of the experience, all visitors reported a deeper, more complex understanding of the cultural meanings embodied in the Surinamese flag.

This design successfully demonstrated that virtual heritage environments can be created with the intention of remaining open-ended, preventing the narrowing of interpretive spaces. Visitors realised through their experiences that what they witnessed was only one of many possible realities, and they acknowledged that they had a choice in how they interacted with and understood the material. Using VR's immersive capabilities to juxtapose multiple constructed realities and highlight their constructed nature to users effectively counters the risk of ideological closure in this reflective design. According to media scholar Janet H. Murray, rather than focusing on a single narrative, digital narratives should emphasise user choice and participation by constructing structures with multiple simultaneous possibilities. When carefully curated, VR can offer what Murray refers to as 'polymorphic narratives', interwoven networks of possibilities unveiled through interaction as opposed to predetermined sequences of events (Murray, 1997). Projects such as these transform immersive media from mere content consumption into places of interpretive co-creation by showcasing heritage in a polyphonic (multi-voiced) manner.

4.4.2. *The Lost Palace (2016)*

In partnership with the digital theatre companies Chomko & Rosier and Uninvited Guests, Historic Royal Palaces (HRP) created the immersive heritage experience The Lost Palace (Clarke, 2016), which faithfully recreates Whitehall Palace, the lar/gest Tudor home, which was destroyed by fire in 1698. Without the use of screens or head-mounted displays, visitors are given handcrafted wooden 'listening devices' embedded with RF positioning chips, actuators and bone conduction speakers (Powell, 2018). Streetlamps, walls and paving stones are transformed into portals for dramatic historical moments, including Henry VIII's secret marriage path, Charles I's execution platform and whispered gunpowder conspiracy plots, as participants traverse contemporary Whitehall streets. GPS and beacon triggers activate 3D binaural soundscapes, vibration pulses and subtle LED signals.

According to Walter Benjamin, aura is the "unique appearance of distance, no matter how close the object may be". Whereas VR immerses users in fully constructed worlds by erasing distance, the Lost Palace restores distance by using sound and haptics to create a sense of proximity despite the original palace being physically absent. The aura is recharged by the space between the visible environment (empty streets) and the sensory input (voices, heartbeats, footsteps), which preserves 'traces of the other'. Despite seeing only modern structures, 82 % of visitors said they had 'a strong feeling of standing where history happened', experiencing a sense of awe or a chill (Historic Royal Palaces, 2017). Rather than digitally recreating facades, the project uses sensory cues in situ to anchor authenticity in Whitehall's physical landscape. Audio-haptic layers generate a timeless 'auratic halo' (Hornecker and Ciolfi, 2019).

Unlike traditional re-enactments, The Lost Palace organises disjointed narrative 'shards'. Rather than providing a chronological

explanation, users gather cryptic fragments, such as the boom of the 1698 fire, the duke's deadly cough or a prostitute's laughter. Benjamin's allegory, whereby meaning arises from assembling historical ruins rather than perfect recreations, is embodied in this design (Benjamin, 1999). Visitors mentally co-create images in the absence of visual elements, exhibiting what Jeffrey refers to as 'imaginative co-production' of digital aura (Jeffrey, 2015b). By encoding multiple voices (those of monarchs, servants and rebels) and encouraging negotiated interpretation through handset controls, HRP enables users to pause, rewind or skip scenes. Depending on their preferences, visitors can prioritise stories that highlight public dissension or royal excess, thereby promoting real-time polysemic memory. Crucially, the roles of curator and audience are blurred in this experience. The participants' wooden handheld device is a 'talisman-like artefact' that signals active mediation and fosters tactile intimacy (Chomko and Rosier, 2016). It is a combination of artefact and interface. Rokeby's 'transforming mirror', in which users' movements are reflected back as historical insights, exemplifies the direct interaction between participants, who can listen through keyholes or press handles to feel the pulse of execution drums (Rokeby and Dodsworth, 1998b). In line with Jeffrey's assertion that co-creation fosters democratisation and authenticity, HRP enlisted community volunteers for voice acting and sought the counsel of playwrights, architects and historians in crafting the scripts for the episodes (Jeffrey, 2015c).

The Lost Palace emphasises intangible sensations — sound, rhythm and touch — over high-definition visuals, illustrating digital aura through evocative absence rather than realistic presence. In contrast, the Surinamese VR exhibition offered visual pluralism in a closed museum gallery. Because the palace is invisible, visitors cannot recognise a single 'master frame'. Instead, they create living allegories by inferring meaning from echoes, vibrations and spatial memory cues. This allegorical structure defies ideological closure.

5. Conclusion

This study revisits the concept of the 'aura', highlighting that in the digital age, authenticity does not arise from original artefacts but rather from viewer–media interactions. According to the 'migratory aura' theory of Latour and Lowe (Latour et al., 2011) and the 'aura in perpetual oscillation' theory of Bolter (Bolter et al., 2006), original authority and emotional resonance can move or redistribute through reproductions or virtual environments. Additionally, Jeffrey contends that digital heritage content can provide viewers with an authenticity that rivals that of originals through democratisation, collaborative production and careful artistic staging. Aura is now a socially and experientially constructed phenomenon that arises through engagement with virtual artefacts rather than being a mystical quality unique to originals. When aura and authenticity in immersive media are acknowledged as performative and migratory characteristics, users become active participants rather than passive consumers.

Drawing upon previous discussions, this study suggests that effective implementation of immersive heritage VR should incorporate multiple and diverse narratives to allow critical reconstruction of historical meanings through comparative engagement, facilitate meaningful interactivity by enabling users to actively explore content and shape their interpretative paths, and adopt a co-authorship model by involving various stakeholders—such as historians, local communities, and indigenous groups—in content creation processes, ensuring that narratives remain inclusive, democratic, and reflective of multiple perspectives beyond singular institutional viewpoints. These approaches collectively transform immersive media into genuinely open spaces for dialogue, interpretation, and critical engagement, moving beyond mere spectacle or passive consumption. To promote a critical understanding of actual histories, virtual heritage should be contextualised by integrating it into thorough historical contexts while openly displaying interpretative ambiguities, source materials and hypothetical reconstruction elements.

These guidelines are an assessment framework developed from the analysis of this study, which can be applied to the future design and evaluation of immersive heritage content. Case studies such as the Lost Palace project and the Surinamese Flag VR exhibition demonstrate how immersive media can maintain interpretative plurality and audience autonomy while providing a high level of realism and emotional immersion. When applied to historical site reproduction, these practical methods of immersive VR can reconstruct multi-layered memories and support empathetic learning. In conclusion, the social effects of immersive VR heritage media depend heavily on design philosophies and implementation techniques; they are not inherently liberating or oppressive. Immersive heritage projects can overcome the constraints of single-narrative storytelling and evolve into inclusive, critically engaging platforms by adopting the principles of plurality, co-authorship, interactivity and transparency suggested in this study. Ultimately, these methods perpetuate and broaden collective memory practices, connecting the past and present in evolving media landscapes, as outlined in Jan Assmann's cultural memory theory. Digital memory does not disappear but changes and evolves with new generations and situations, demonstrating how historical engagement is evolving thanks to immersive media.

Declaration of competing interest

The author declares no competing financial, personal, or professional interests that could influence or bias the content of this manuscript. There are no conflicts of interest to report.

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